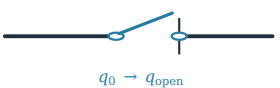
FROM PHYSICAL INTERVENTION TO CAUSAL PMU REPORT

Restoration returns the network operator to base while dynamic and measurement histories continue.

01 PHYSICAL OPERATORS

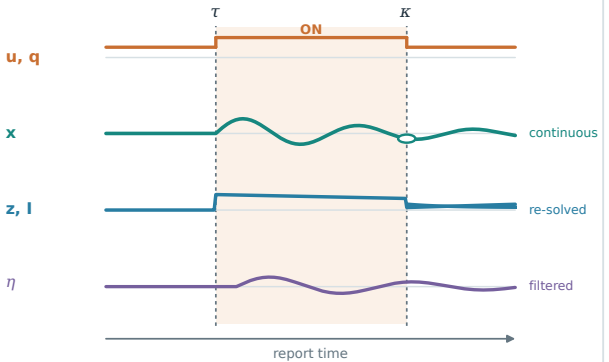
LOAD $P, Q \mapsto (1+a)(P, Q)$

LINE



generic operator symbols; not bus locations

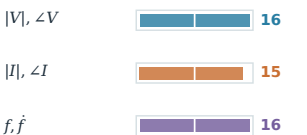
02 CAUSAL RESPONSE THROUGH RESTORATION



OFF restores the operator; it does not reset x or the filter state.

illustrative trajectories — not measured records

03 PMU REPORT



$y_k \in \mathbb{R}^{47}$

30 reports/s · causal prefix only

u, q return to base · x, η persist · z and terminal I are re-solved